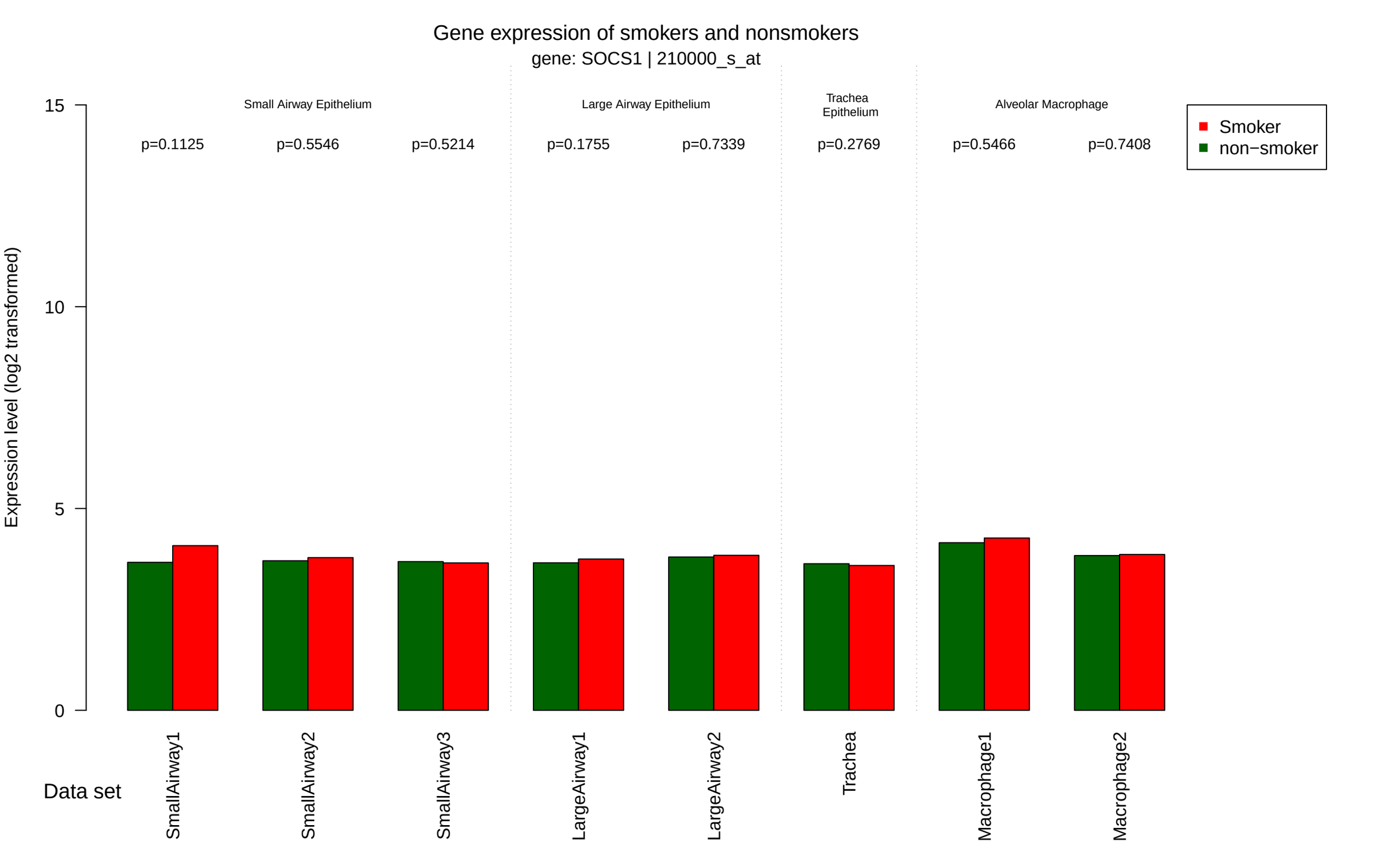
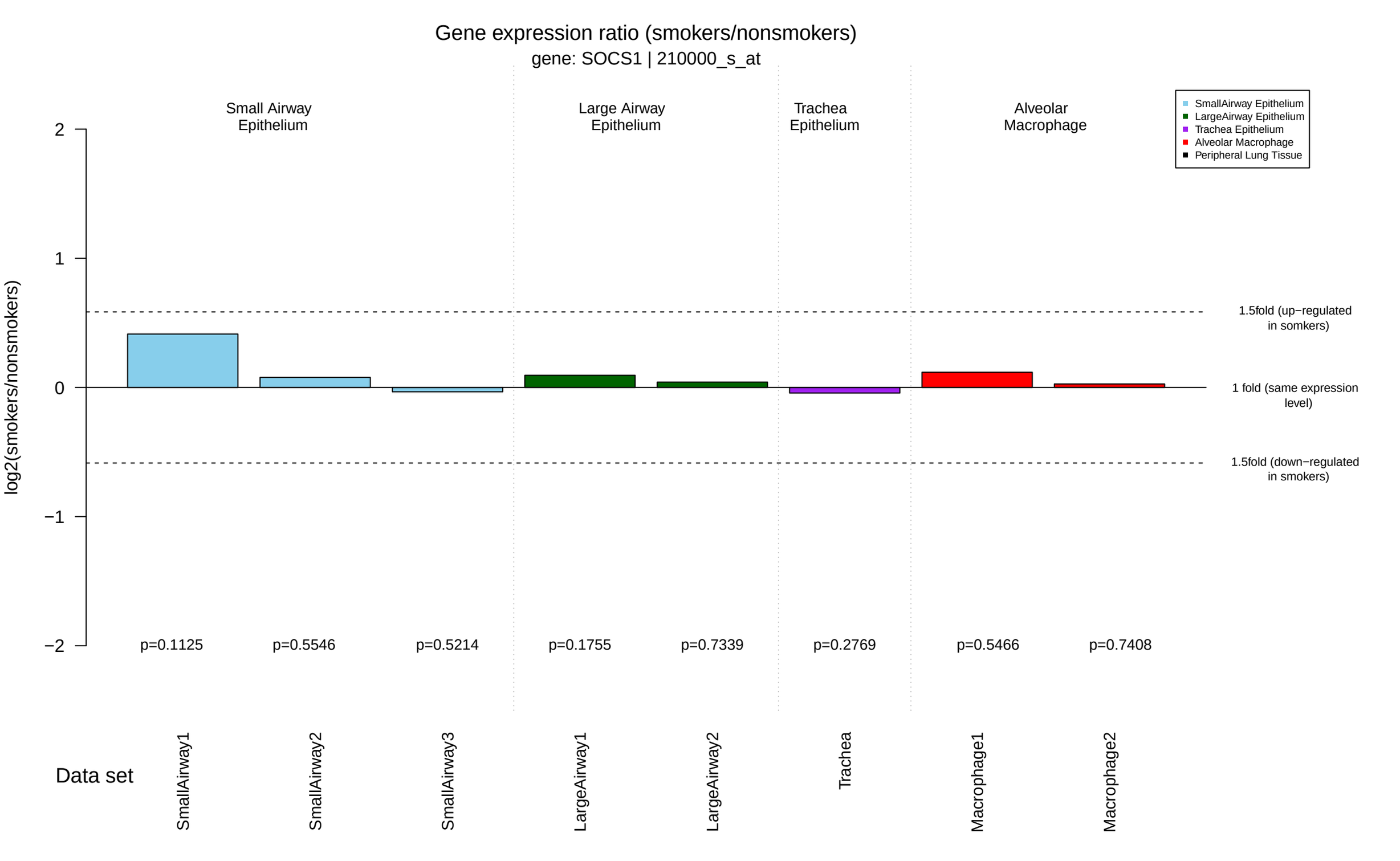
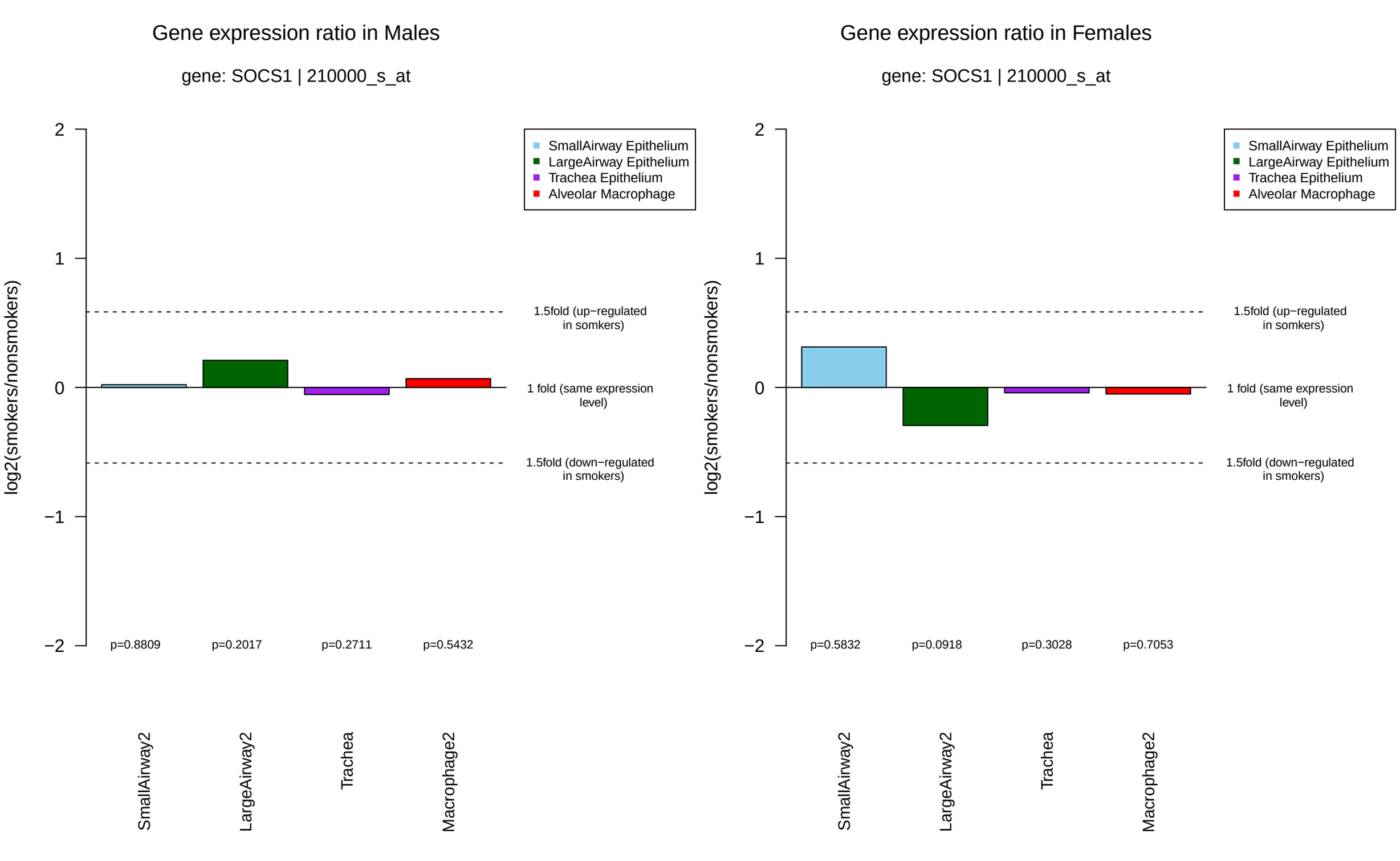
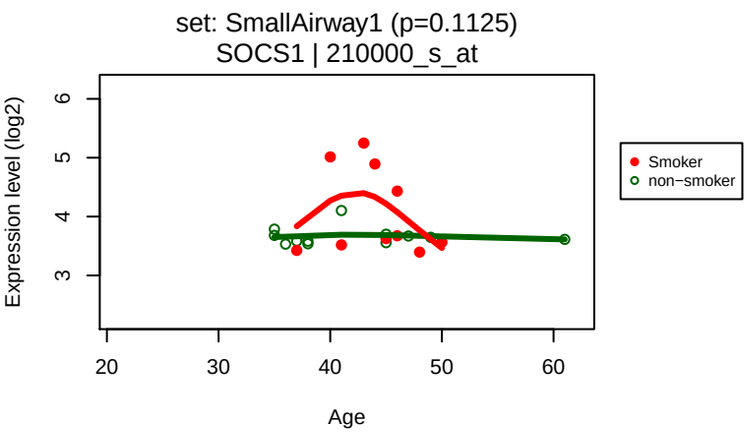
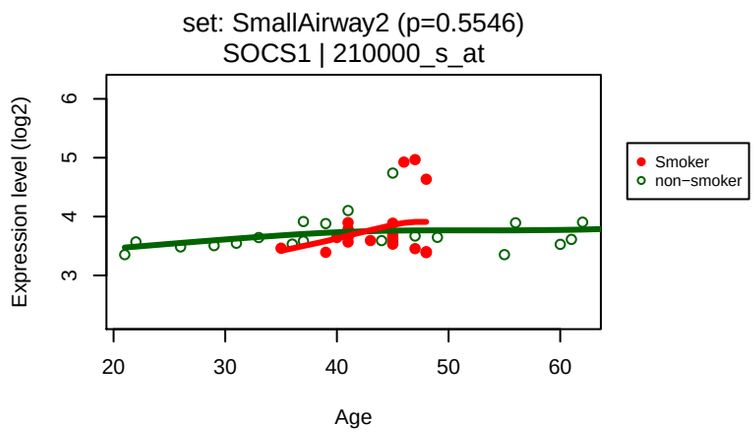
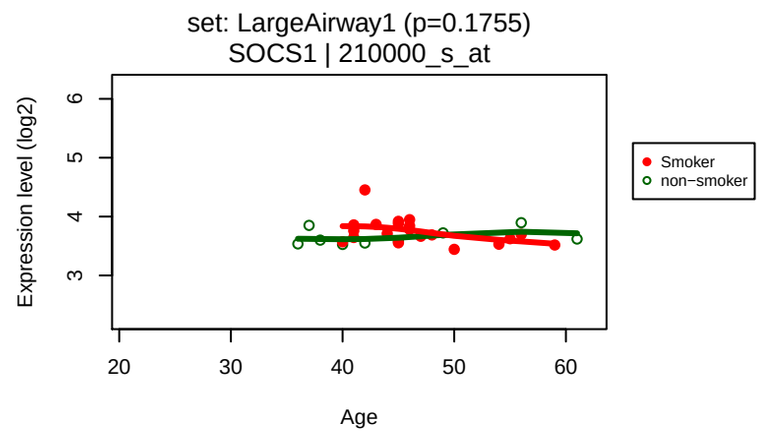

Gene expression of smokers and nonsmokers
gene: SOCS1 | 210000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1125
p=0.5546
p=0.5214
p=0.1755
p=0.7339
p=0.2769
p=0.5466
p=0.7408
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

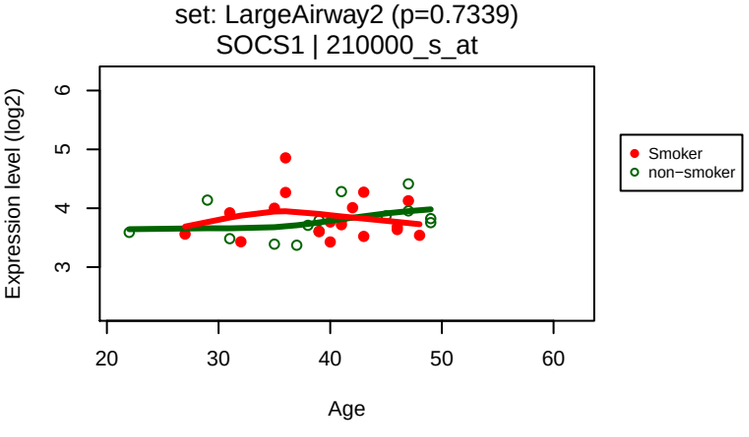

Gene expression ratio (smokers/nonsmokers)
gene: SOCS1 | 210000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1125
p=0.5546
p=0.5214
p=0.1755
p=0.7339
p=0.2769
p=0.5466
p=0.7408
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

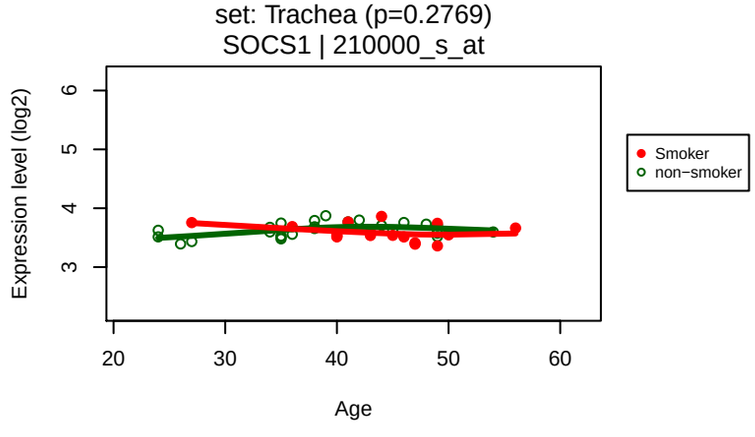

Gene expression ratio in Males
gene: SOCS1 | 210000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8809
p=0.2017
p=0.2711
p=0.5432
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS1 | 210000_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5832
p=0.0918
p=0.3028
p=0.7053
SmallAirway2
LargeAirway2
Trachea
Macrophage2

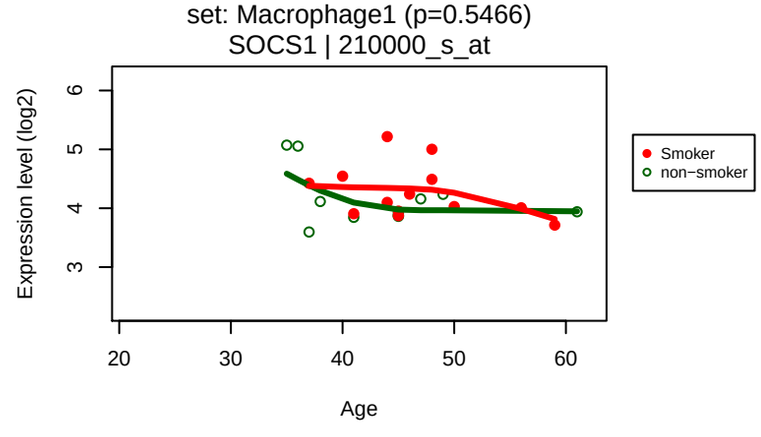

set: SmallAirway1 (p=0.1125)
SOCS1 | 210000_s_at
Expression level (log2)
Age
Smoker
non−smoker

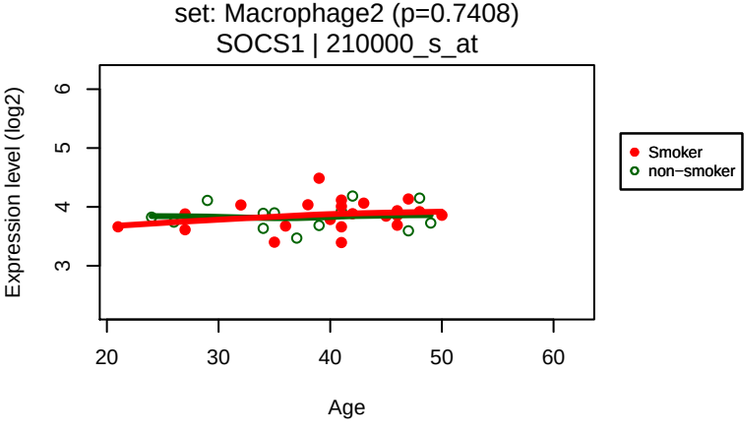

set: SmallAirway2 (p=0.5546)
SOCS1 | 210000_s_at
Expression level (log2)
Age
Smoker
non−smoker

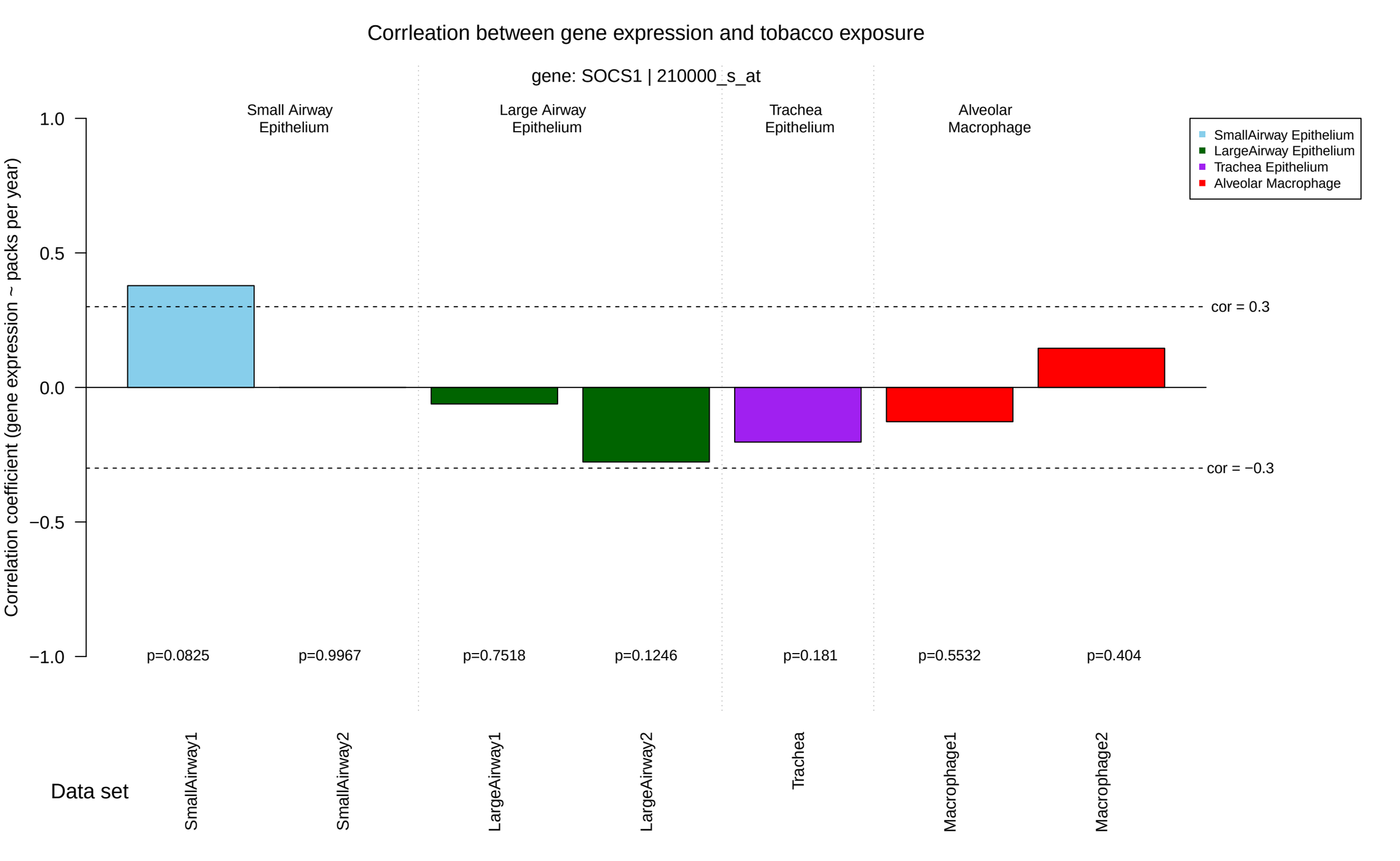

set: LargeAirway1 (p=0.1755)
SOCS1 | 210000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7339)
SOCS1 | 210000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2769)
SOCS1 | 210000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5466)
SOCS1 | 210000_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7408)
SOCS1 | 210000_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS1 | 210000_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0825
p=0.9967
p=0.7518
p=0.1246
p=0.181
p=0.5532
p=0.404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2